

بافت

1- Synthesis of proteins that are used in mitochondria is the function of سوال

SER

RER

Attached Ribosomes

✓ Free Ribosomes

2- Which one of these organelles can replicate before the cell division stage? سوال

RER

Ribosome

✓ mitochondria

SER

3- سوال The cytoskeleton is a system of ____

Glycolipids

✓ Proteins

Proteoglycans

Carbohydrates

4- Which of the following cellular components retains the greatest structural integrity throughout the cell cycle?



Nucleolus

Centriole

Golgi complex

Nuclear envelope

5- Which of the following terms is applied to the smallest chromosome subunit in eukaryotic cells? سوال

Chromatid

Euchromatin

heterochromatin



Nucleosome

6- Which of this structure DONOT enclose by membrane? سوال



Proteasome

Peroxisomes

Lysosomes

Endosomes

7- Synthesis of Histones occurs in.....

G2



S

G0

G1

8- In the cell membrane, which one can be used as channel and receptor? سوال

Peripheral protein

Cholesterol

Phospholipid

✓ Integral protein

9- سوال Arrangement of 9×3 microtubules in eukaryotic cell is referred to as.....

Flagella

Microvilli

✓ Centriole

Cillia

Physiology

10- In which type of intercellular communication, the products of cells diffuse in the ECF to affect neighboring cells

Autocrine

✓ Paracrine

Neural

Endocrine

11- Glucose transport through SGLT1 is a type of

simple diffusion

primary active transport

✓ secondary active transport

facilitated diffusion

12- Which of the following component controls much of the fluidity of the membrane?

Sphingolipids

✓ Cholesterol

Phospholipids

Integral proteins

13- Which of the below transport mechanism needs ATP?

Bulk flow

Osmosis

Solvent drag

✓ Exocytosis

14- Which of the following is the most important factor in determining osmotic pressure?

Kinetic motion of matter

✓ Number of osmotic particles

Velocity of osmotic particles

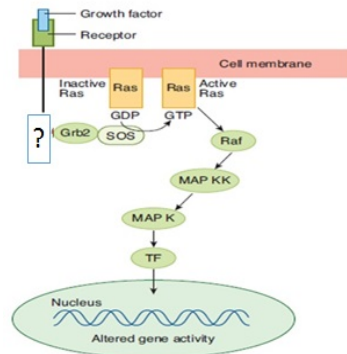
Mass of osmotic particles

15- A hypotonic solution has been injected to a 25 years old man. What happens in the cells of his body?



- Cell volume increases.
- Cell osmolarity increases.
- Shriveled cells are seen.
- No change is seen.

16- What should be written instead of question mark?



- Tyrosine kinase
- JAK
- guanylate cyclase
- adenylate cyclase

17- Which type of glucose transporters are found in pancreatic β cell glucose sensor?

- GLUT1
- ✓ GLUT2
- GLUT3
- GLUT4

18- Which ion does the membrane have the most permeability to at resting state?

- Na^+
- ✓ K^+
- Cl^-
- Ca^{2+}

19- Which of the following is not correct about "internal environment"?

- Fluid outside the cells
- ✓ 2/3 volume of fluids in body
- Containing ions & nutrients needed for cellular life
- Containing waste substances

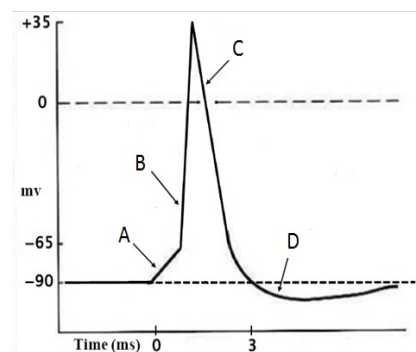
20- Electrogenic nature is a main function of

- Na^+ / H^+ exchanger
- Leak channels
- ✓ Na^+ / K^+ pump
- SGLT 1

21- Which of the following points to the leak channels properties?

- ✓ Being always open
- Having gate(s)
- Undergoing conformational changes
- Binding with molecules or ions that are to be transported

22- According to this figure, which part of action potential shows “overshoot”?



B

D

A

C



23- Which factors are considered in Goldman equation?

- ☐ Concentration and electrical potential
- ☒ Concentration and permeability
- ☐ Permeability and number of channels
- ☐ Capacity and electrical potential

24- Which second messenger activates protein kinase A?

- ☒ cAMP
- ☐ Ca^{2+}
- ☐ DAG
- ☐ IP3

25- Calculate the Nernst potential for hypothetical X^{+1} ions.

Intracellular concentration: 2 mEq/Lit Extracellular concentration: 200 mEq/Lit

$$\text{Log } 100 = 2 \text{Log } 0.01 = -2$$

- ☐ +61 mv
- ☐ -61 mv
- ☐ -122 mv
- ☒ +122 mv

26- Refractory period is caused because of

- subthreshold potentials
- passage of electricity through the membrane
- all or nothing principle
- ✓ inactivated voltage gated Na^+ channels

Biochemistry

27- Which of the following vitamins provide the coenzyme for γ -carboxylation reactions?

- ✓ Menaquinone
- Tochopherol
- Biotin
- Pyridoxine

28- Which of the following enzymes is required for end to end joining of DNA?

- RNA polymerase
- Restriction endonuclease
- ✓ DNA ligase
- DNA polymerase

29- A tripeptide has?

- ☐ 3 amino acids and 4 peptide bonds
- ☐ 3 amino acids and 1 peptide bonds
- ☒ 3 amino acids and 2 peptide bonds
- ☐ 3 amino acids and 3 peptide bonds

30- Which substance is produced in every dehydration synthesis reaction?

- ☒ Water
- ☐ Fat
- ☐ Protein
- ☐ Carbohydrate

31- A buffer solution is 0.080 M in lactic acid ($pK_a=3.74$) and 0.070 M in sodium lactate. The pH of the solution is?

- ☒ 3.68
- ☐ 4.62
- ☐ 4.16
- ☐ 2.86

32- Which of the following mutations would produce a minor thalassemia?

- ☐ Altered RNA processing at both β -globin loci
- ☐ Oxidation of heme groups to produce methemoglobin
- ☒ Deletion of one β -globin locus
- ☐ Deletion of one α -globin locus

33- All of the following statements regarding fibrous proteins are true except?

Keratin and collagen are the best examples



Hemoglobin is the best example

It is in rod or wire like shape

It provides structural support for cells and tissues

34- The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?

2.5

4.6

6.9



8.5

35- Which of the following lac operon genes has constitutive expression?



i

p

c

z

36- Which of the following statements about Michaelis-Menten constant is correct?

The K_m of an enzyme is unchanged when the enzyme concentration is increased.



A low K_m indicates a high affinity of an enzyme for its substrate.

A high K_m indicates a high affinity of an enzyme for its substrate.

The K_m of an enzyme increases when the enzyme concentration is increased.

37- A patient gets abdominal pains and cramps, diarrhea and gas in intestines after consuming milk and dairy products. His problem is related to a defect in which sugar and enzyme below?

- ☒ Lactose-lactase
- Galactose-galactokinase
- Fructose-fructokinase
- Glucose-hexokinase

38- How many vitamins are classified as fat-soluble?

- 13
- ☒ 4
- 2
- 9

39- Peptide bond is a?

- Hydrogen bond
- Metallic bond
- Ionic bond
- ☒ Covalent bond

40- For octanoic acid, $pK_a = 4.89$. Which pH range would be the best buffering region for this acid?

- ✓ 3.89-5.89
- 3.5-4.5
- 5.5-7.5
- 3.89-4.89

41- An example of glycerophospholipid involved in cell signaling is?

- ✓ Phosphatidylinositol
- Cardiolipin
- Phosphatidylcholine
- Phosphatidic acid

42- Dietary fats after absorption appear in the circulation as?

- ✓ Chylomicron
- VLDL
- HDL
- LDL

43- All the following compounds possess a chiral carbon, except?

- Mannose
- ✓ Dihydroxyacetone
- Glyceraldehyde
- Ribose

44- Which of these operons is anabolic?

- ☐ Ara
- ☒ Trp
- ☐ Lac
- ☐ Phe

45- When $S=K_m$, the V is equal to.....?

- ☒ $\frac{1}{2} V_{max}$
- ☐ $\frac{2}{3} V_{max}$
- ☐ $2 V_{max}$
- ☐ $1 V_{max}$

46- Which of the following are anomeric isomers?

- ☐ D-glucose and L-fructose
- ☒ α -D-Glucose and β -D-glucose
- ☐ D-Galactose and D-mannose
- ☐ D-Glucose and D-galactose

47- Tetracycline prevents synthesis of polypeptide by

- ☒ Preventing binding of aminoacyl tRNA
- ☐ Releasing peptides from mRNA-tRNA complex
- ☐ Blocking mRNA formation from DNA
- ☐ Competing with mRNA for ribosomal binding sites

48- The initiation site for transcription is recognized by

- ☐ Rho factor
- ☒ Sigma factor
- ☐ β -Subunit of DNA-dependent RNA polymerase
- ☐ α -Subunit of DNA-dependent RNA polymerase

49- Which of the substances below is primarily found in tendons?

- ☐ Fibrillin
- ☐ Fibrin
- ☒ Collagen
- ☐ Troponin

50- Splice sites are present in

- ☐ Prokaryotic mRNA
- ☒ Eukaryotic hnRNA
- ☐ Prokaryotic hnRNA
- ☐ Eukaryotic mRNA

51- Which of the following is correctly matched with the vitamin it is derived from?

- ☐ FADH₂-Vitamin B₃
- ☐ TPP-Vitamin C
- ☐ NADH-Vitamin B₂
- ☒ Pyridoxal phosphate-Vitamin B₆

52- Which vitamin is a methyl group acceptor?

- ☒ Vitamin B12
- ☐ Coenzyme A
- ☐ Lipoic acid
- ☐ Vitamin D

53- Which of the following can act as a feedback inhibitor of an enzyme?

- ☒ End product
- ☐ Substrate
- ☐ Enzyme
- ☐ External factors

54- Which of the following is not a clinical outcome of niacin deficiency?

- ☐ Diarrhea
- ☐ Dermatitis
- ☐ Dementia
- ☒ Night blindness

55- Dipalmitoyl lecithin acts as?

- ☐ Anti-ketogenic compound
- ☐ Second messenger for hormones
- ☐ Platelet activating factor
- ☒ Lung surfactant

56- Which of the following is a homo-polysaccharide?

- Ascorbic acid
- Hyaluronic acid
- ✓ Cellulose
- Heparin

57- Which of the following statements is true?

- ✓ G-C bonds are much more resistant to denaturation than A-T rich regions
- The common form of DNA is left-handed
- The two strands of DNA run parallel to each other
- The template strand matches the sequence of the RNA transcript

58- The oxygen carrier of muscle is the globular protein myoglobin. Which one of the following amino acids is highly likely to be localized within the interior of the molecule?

- Arginine
- Glutamic acid
- ✓ Valine
- Aspartic acid

59- Which of these is a vitamin A precursor?

- Thiamine
- Cobalamin
- ✓ Beta-carotene
- Pyridoxine

60- Which of the following is an essential fatty acid?

Palmitoleic acid

Oleic acid

Stearic acid

✓ Linoleic acid

61- Which of the following statements about solutions of amino acids at physiologic pH is true?

Some amino acids contain only positive charges

✓ All amino acids contain both positive and negative charges

All amino acids contain positively charged side chains

All amino acids contain negatively charged side chains

62- The enzyme which hydrolyses starch to maltose is?

Lactase

Sucrose

Protease

✓ Amylase

63- During synthesis of mature collagen fiber, which one of the following steps would occur within the fibroblast?

Formation of covalent cross-links between molecules

Hydrolysis of procollagen to form collagen

Glycosylation of proline residues

✓ Formation of a triple helix

64- Toothpaste and sugar-free gum are probably sweetened with a sugar alcohol made by reducing the aldehyde carbon of xylose to an alcohol. The resulting compound would be named?

Mannose



Xylitol

Sorbitol

Arabinose

65- Zymogen or proenzyme is a?

Hormone



Enzyme precursor

Modulator

Vitamin

66- In which of the following disaccharides the glycosidic linkage is in between anomeric carbons of both monosaccharides?

Lactose



Sucrose

Cellobiose

Maltose

67- Which of the following fatty acid has the least melting point?

Arachidonic acid

Palmitic acid

✓ Eicosapentaenoic acid

Stearic acid

68- A coiled peptide chain held in place by hydrogen bonding between peptide bonds in the same chain (not between two chains) is?

β -pleated sheets

Tertiary structure

Primary structure

✓ α -helix

69- Which of the following is the active form of pantothenic acid?

Vitamin B5

TPP

PLP

✓ Coenzyme A

70- HDL is synthesized in?

Adipose tissue

Intestine

Liver

✓ Liver and intestine

71- The effect of reversible competitive inhibitor can be canceled out by?

- Increasing the pH
- Increasing the temperature
- ✓ Increasing the substrate concentration
- Increasing the product concentration

72- Xeroderma pigmentosum results from a defect in?

- DNA polymerase III
- DNA polymerase I
- DNA ligase
- ✓ uvr ABC exonuclease

73- Which one of the following nucleic acids has a left-handed helix?

- M-RNA
- T-RNA
- ✓ Z-DNA
- A-DNA

74- Okazaki fragments are formed during the synthesis of?

- tRNA
- rRNA
- mRNA
- ✓ DNA

75- Compounds that are needed for enzymes to function properly are?

Steroids

Heavy metals

Buffers

✓ Vitamins

76- A 60-year old female presents with severe back pain for the past week, she is diagnosed with osteoporosis, a condition resulting from calcium depletion in bones. As treatment which of the following vitamins can be prescribed?

Vit A

✓ Vit D

Vit K

Vit E

77- Which of the following belong to glycolipids?

✓ Cerebrosides

N-acetylgalactosamines

N-acetylglucosamines

Xyloses

78- mRNA is complementary to the nucleotide sequence of?

- ☐ tRNA
- ☒ Coding strand
- ☐ Template strand
- ☐ Ribosomal RNA

79- Ciprofloxacin inhibits?

- ☒ DNA gyrase
- ☐ DNA polymerase I
- ☐ DNA topoisomerase I
- ☐ DNA polymerase III

80- Deficiency of lysyl hydroxylase causes syndrome characterized by hypermobile joints and skin abnormalities.

- ☐ Marphan
- ☒ Ehlers-Danlos
- ☐ Scurvy
- ☐ X-fragile

81- Which vitamin(s) play important roles in the metabolism of branched-chain amino acids and odd carbon fatty acids?

- ☐ Pyridoxine
- ☐ Thiamine
- ☒ Biotin
- ☐ Folic acid

82- The structure of tRNA appears like a?

Helix

Coil

Hair pin

✓ Clover leaf

83- The ability of hemoglobin to serve as an effective transporter of oxygen and carbon dioxide between lungs and tissues is explained by which of the following properties?

Hemoglobin binds proportionately more oxygen at low oxygen tension than does myoglobin

Hemoglobin utilizes oxidized ferric iron to bind oxygen, in contrast to the ferrous ion of myoglobin

The isolated heme group with ferrous iron binds oxygen much more avidly than carbon dioxide

✓ In contrast to myoglobin, hemoglobin exhibits greater changes in secondary and tertiary structure after oxygen binding

84- The lipoprotein richest in cholesterol is?

HDL

Chylomicron

✓ LDL

VLDL